

Log Likelihood Explainer

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What Likelihood Is

Likelihood is the probability of observing a particular outcome, assuming a fixed model of how the data is generated. For example, if we assume a fair coin flip (i.e., each flip has a 0.5 probability of landing heads), then observing two heads in two flips has a likelihood of $0.5 \times 0.5 = 0.25$. However, if we instead assume that the probability of heads is 0.3, then the likelihood of observing two heads becomes $0.3 \times 0.3 = 0.09$. For *reasons* we say that a Likelihood of no observation is always 1. Importantly, this is not the same as the probability of the model given the observation. That would require prior beliefs about the model itself, which we do not assume. Confusing these two quantities is known as the prosecutor’s fallacy—mistaking $P(\text{model} \mid \text{data})$ for $P(\text{data} \mid \text{model})$.

Log Likelihood

Log likelihood is simply the likelihood function wrapped in the logarithm function. The base of the logarithm is an open question to me right now, but it seems that that decision is often dependent upon the field of study. This transformation has two main advantages. First, the logarithm is a strictly increasing function, so maximizing the log likelihood is equivalent to maximizing the likelihood itself. Second, it simplifies arithmetic: ratios of probabilities become differences, $\log(a/b) = \log a - \log b$, and joint probabilities become sums, $\log(a \times b) = \log a + \log b$. This makes optimization and numerical computation much easier. As well the concept of no observation is much more intuitive, while the likelihood of not observing something is set to 1, the log likelihood of 1 becomes 0.